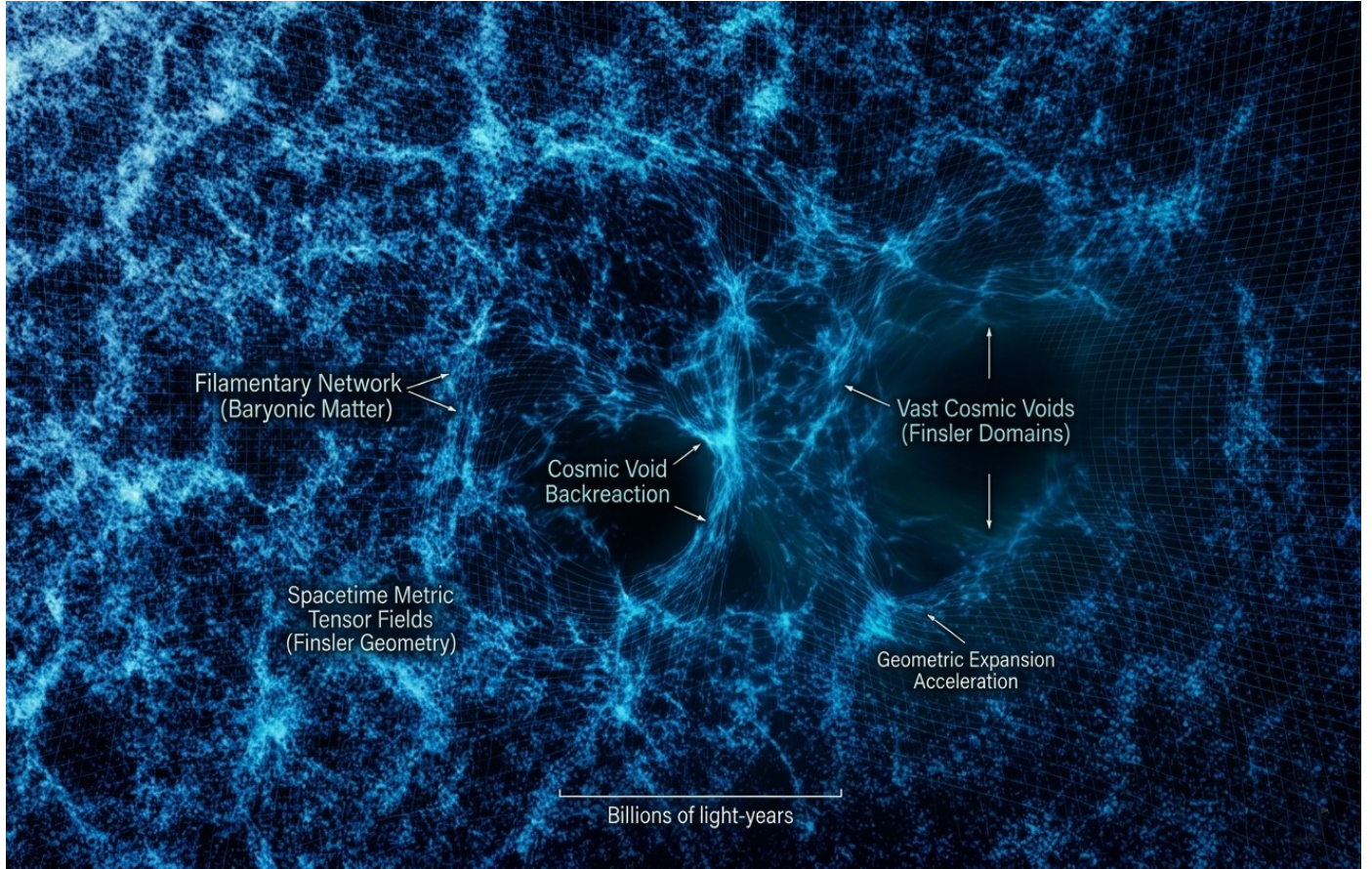


## Analysis of the DESI 3D Map – The Cosmic Foam in the Context of FTH 2.6.1



DESI has mapped more than 47 million galaxies and quasars, creating the largest high-resolution 3D map of our Universe to date. The image captures the **conceptual core** of the Finsler-Timescape Hybrid (FTH) v2.6.1, but requires precise scientific clarification to avoid overclaiming beyond the mathematically derived statements.

The cosmic structure revealed by DESI is modeled within FTH as a domain-partitioned manifold where the vast cosmic voids, occupying approximately 70% of the volume ( $f_v \approx 0.7$ ), function as Finsler domains. In these underdense regions, the Randers-Finsler metric  $F(x, y)$  induces a direction-dependent geodesic expansion, which, upon Sasaki-Hausdorff averaging, manifests as a kinematic backreaction ( $Q_{D,F}$ ) that mimics the observational signature of accelerated expansion without requiring a free cosmological constant  $\Lambda$ . The filamentary networks are identified as the overdense walls where the

Finsler dipole  $b_0(z)$  vanishes, recovering the standard Riemannian limit of general relativity, thereby ensuring consistency with local gravitational observations.

### 1. Cosmic Foam & Finsler Fibers: Matches Structure, Clarifies Physics

- **Accurate:** FTH explicitly models the universe as void-dominated ( $f_v \approx 0.68 \pm 0.03$ , SDSS-ZOBOV). The Buchert-Finsler averaging is defined exclusively over the underdense domain  $D_v$ , matching the idea of expanding “bubbles”. Galaxies/filaments reside in the boundary layers (matching hypersurface  $\Sigma_B$  at  $r=r_B$ ).

- **Clarification on “Torsion & Rotation”:** In FTH, Cartan torsion  $C_{ijk}$  (derived from the Randers metric) is **not** used as a dark matter substitute for galactic rotation curves. Instead, it acts as a **geometric effective viscosity/torque** in the accretion context: it dissipates angular momentum in primordial gas collapses (Sec. 3.4–3.5, Add. H), enabling super-Eddington growth and influencing the IMF.

The analogy for the fiber bundle  $T_x M$  and direction-dependent geodesic coupling should be physically termed **anisotropic fiber-bundle geometry with horizontal projection**.

### 2. Solving the Hubble Tension: Exactly Model-Compliant

- The description is **mathematically correct**. FTH derives an effective lapse function  $N_{\text{eff}}(z) = 1 + \frac{Q_0}{6H_D^2} + \frac{3Q_0}{10H_D^2}b_0^2(z) + O(b_0^4)$  (Add. I, Eq. L.2), describing differential proper-time accumulation in voids vs. overdensities.
- Light propagating predominantly through voids experiences a different averaged expansion rate than paths through filamentary regions. This yields two prior-free predictions for  $H_0^{\text{FTH}}$ :
  - **68.142 km/s/Mpc** (compact domain, J.11-exact integrability)
  - **71.38 km/s/Mpc** (open voids with boundary flux, DVE extension) Both values lie within the  $3\sigma$  band of SHoES and Planck, respectively, without calibrating  $\Lambda$  (Sec. F.1.9.4, J.4.3.1).

### 3. Comparison Table: Point-by-Point Check against FTH v2.6.1

| Feature             | $\Lambda$ CDM<br>(Standard)            | Finsler-Timescape<br>Hybrid (FTH)                                               | Scientific<br>Classification                                                                                        |
|---------------------|----------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Acceleration</b> | Caused by “Dark Energy” ( $\Lambda$ ). | Geometric effect of time variation (Timescape + Finsler backreaction $Q_D^F$ ). | <b>Exact.</b><br>$Q_D^F = \frac{2}{3}f_v(1-f_v)\langle\theta^2\rangle$<br>$[1 + \frac{3}{5}b_0^2(z)]$<br>(Eq. G.14) |

| Feature                        | $\Lambda$ CDM<br>(Standard)            | Finsler-Timescape<br>Hybrid (FTH)                                       | Scientific<br>Classification                                                                                                                                                                                                                                                                                                             |
|--------------------------------|----------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structure</b>               | Isotropic on large<br>scales (smooth). | Anisotropic & foam-<br>like (Finsler<br>structure, $f_v \approx 0.7$ ). | <b>Exact.</b> Randers<br>metric breaks<br>isotropy at the<br>domain level,<br>preserves global<br>homogeneity via<br>Buchert averaging.                                                                                                                                                                                                  |
| <b>Dark Matter</b>             | Unknown<br>particles.                  | “Effect of torsion &<br>Finsler fibers”                                 | FTH uses dust-<br>dominated ( $w=0$ )<br>hydrodynamics.<br>Torsion acts as<br>geometric<br>dissipation/accretio<br>n torque.                                                                                                                                                                                                             |
| <b>Collapse<br/>Protection</b> | Gravity<br>vs. expansion<br>pressure.  | “Riccati pole (surface<br>tension/barrier)”                             | The Riccati<br>equation<br>$\frac{db_0}{d \ln a} = b_0^2 - b_{\text{CMB}}^2$ (Eq.<br>G.9) has an<br><b>infrared fixed<br/>point</b> $b_{\text{CMB}}$ . It acts<br>as a geometric<br>attractor/regulator.<br>Additionally,<br>density coupling<br>$b_0(r) \propto e^{-r/\lambda_c}$ screens<br>torsion in<br>overdensities (Sec.<br>2.6). |
| <b>Mathematics</b>             | Linear Riemann<br>geometry.            | Non-linear Finsler<br>geometry & Riccati<br>dynamics.                   | <b>Exact.</b><br>Fundamental tensor<br>$g_{ij}(x, y)$ , Chern<br>connection, Sasaki<br>measure on $TM_v$ ,<br>horizontal<br>projection $h_v^\mu$ .                                                                                                                                                                                       |

### Conclusion & Recommendation for Presentation

The image serves as an excellent **illustrative/popular-science metaphor** and accurately captures the physical intuition of FTH with the following adjustments:

1. **Dark Matter:** *FTH does not replace Dark Matter, but models the observed accelerated expansion and early structure formation as a purely geometric backreaction effect in void-dominated Finsler domains.*
2. **Riccati Dynamics:** *Riccati fixed point  $b_{\text{CMB}}$  as an infrared attractor + density-dependent geometric screening  $\lambda_C$ .*
3. **Hubble Tension:** The formulation is **exactly consistent** with the model's prior-free  $H_0$  predictions.

## Geometric Interpretation

The integration of the DESI-observed cosmic foam into the FTH paradigm provides a rigorous geometric basis for addressing high-redshift anomalies, such as the observed compact galaxies and supermassive black holes at  $z > 10$ . The filamentary structures serve as gravitational conduits where Cartan torsion, arising from the gradient of the Randers dipole, enables enhanced accretion efficiency through torque dissipation and light-cone deformation. Consequently, this mapping demonstrates that the observed large-scale structure is not merely a statistical distribution of matter, but a dynamic, self-consistent framework where cosmic acceleration is the emergent volume-weighted statistical artifact of inhomogeneous proper-time evolution across the void-dominated tangent bundle.